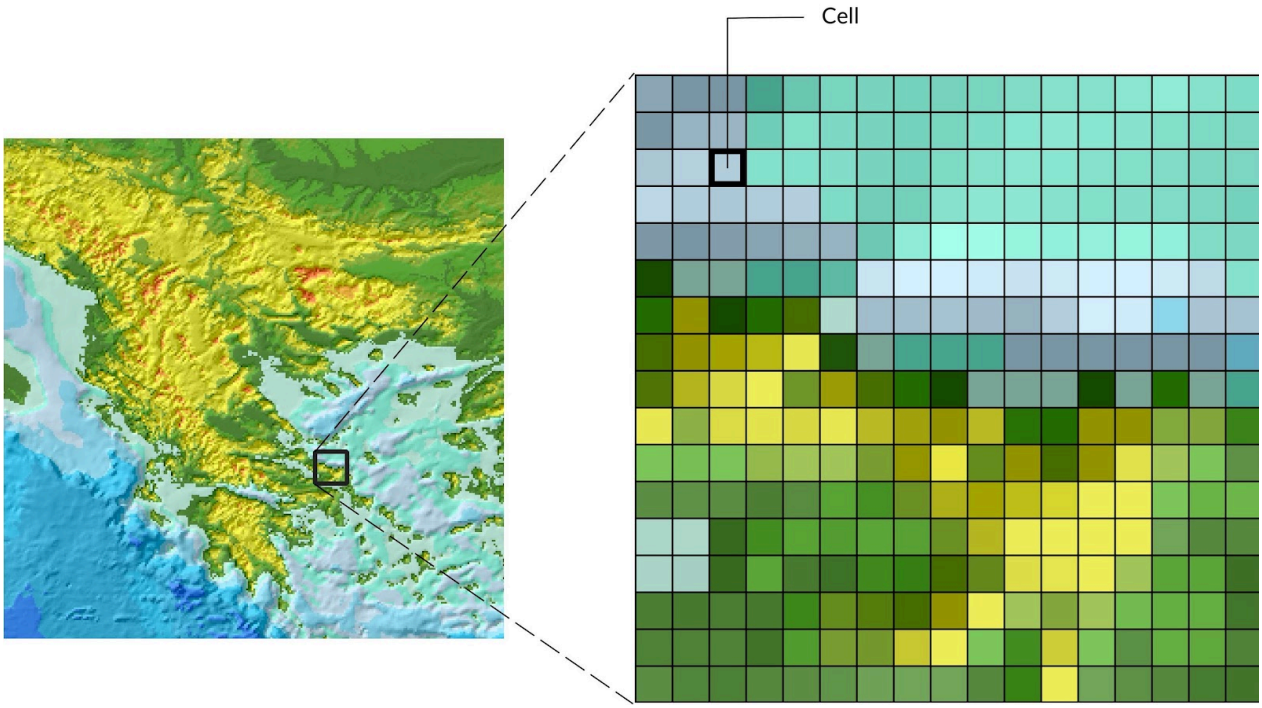
2.1 Raster model

In GIS, you work with information that has both a spatial component (defined by coordinates) and a non-spatial component (attribute data). This information can be structured technically in different ways. The two most commonly used data models in GIS are the **raster model** and the **vector model**. They differ in how they store the spatial component and in what kind of attribute data can be linked to it. We start with the raster model.

Rasters define geographic information by means of a grid of cells (or pixels) organised in rows and columns. Below, we take a short look at some characteristics and give some examples.

Grid of cells

In a raster file, geographic space is divided into a matrix of cells, each covering a small part of that space. Each cell is implicitly given coordinates. The real-world size of one cell (e.g. 10 by 10 metres) is called the **spatial resolution**; it differs from raster file to raster file. Within one raster, however, all cells always have the **same size and shape**.



Data: SRTM shaded relief, USGS (public domain).

Raster files are easy to recognise: at first sight you cannot always see the individual cells, but if you zoom in far enough, the matrix of cells becomes clearly visible.

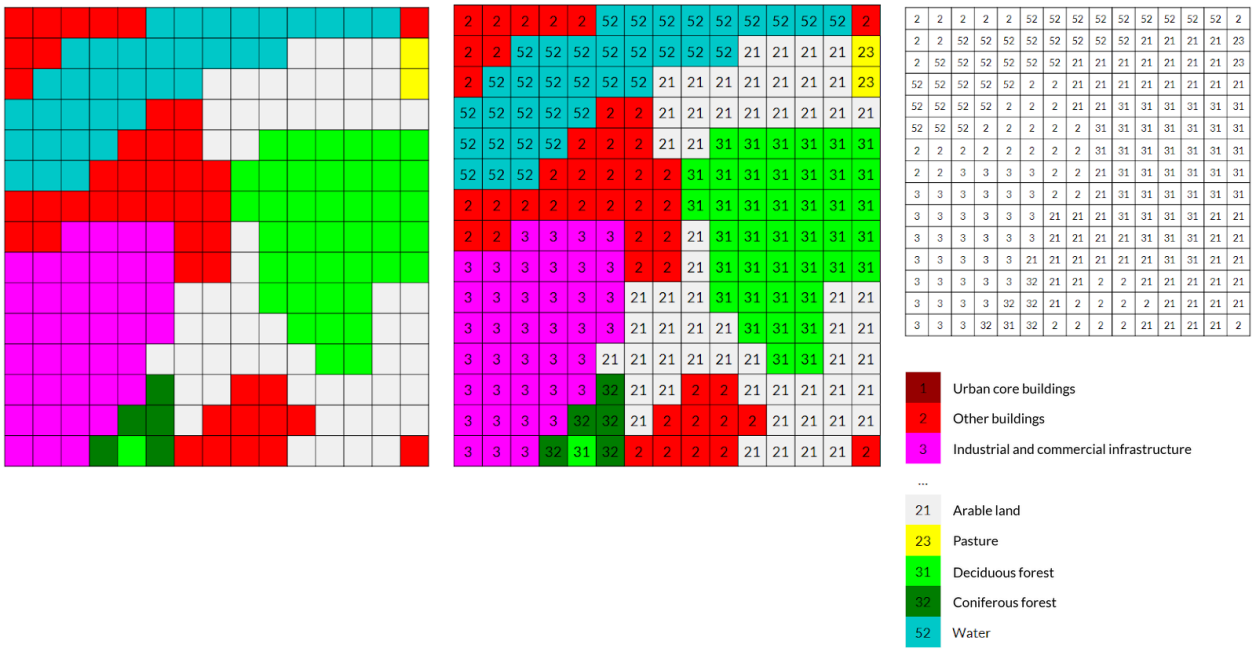
Numeric values

Each cell can be given one or more numeric values, which apply to the whole area covered by the cell. The unit of these values differs from raster to raster, and the options are almost unlimited. Common examples are rasters in which each cell contains the elevation or the temperature.

Although the values of raster cells are always numeric, you can also use a raster file to show textual data, by means of a **code system** and a **legend**. Take the example of the official land use map of the Flemish

government. Each cell of this raster map contains a number as its value. The legend that comes with it explains that a cell with value 1 corresponds to 'urban core buildings', 2 to 'other buildings', 3 to 'industrial and commercial infrastructure', etc. In this way, the values in the cells are numeric, but their meaning is not.

The example below makes the logic of this system clear:



> See also
>
> How to look up the values of raster cells in QGIS is explained on Cell values: histogram and unique values.

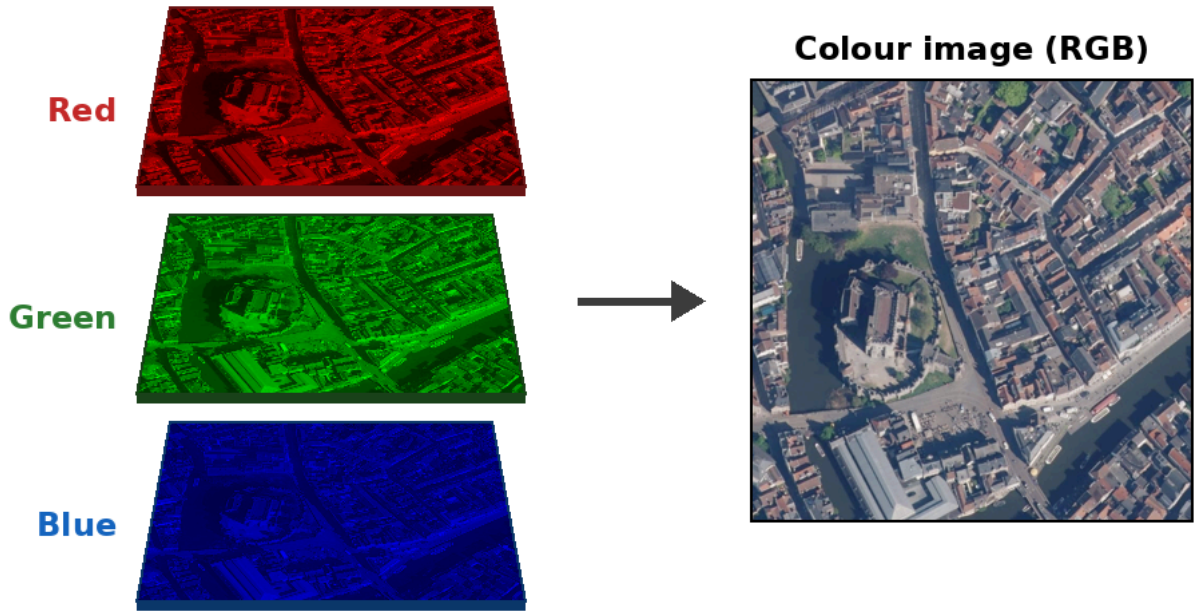
Single-band and multi-band

In the example above, we looked at a simple raster file in which each pixel was given at most one numeric value. Such images are also called *single-band images*.

However, a raster image can also consist of several image layers (*image bands*). In that case, each pixel has one value per image band. All bands of one raster share exactly the same grid: the same number of rows and columns, the same cell size and the same position. You can picture them as sheets stacked on top of each other, in which every cell lies exactly above the corresponding cell in the other bands; only the values stored in the cells differ from band to band. Colour scans (e.g. a scanned historical map) and digital colour photos (e.g. aerial photos) are examples of such *multi-band* raster files. When an image is scanned or photographed, it is converted into a digital raster of pixels, and the scanner or camera gives each pixel a colour in the form of a numeric code. There are different ways to express colour numerically. A common one is **'RGB'** (red - green - blue): here each pixel is given not one value but three, containing the amounts of red, green and blue that together form the colour.

Warning

A scanned map contains colours, not the values the map shows. If you scan a paper elevation map, for example, the cells of your digital raster will not contain the elevations themselves (a scanner cannot know these), but the numeric codes of the colours used on the original map.



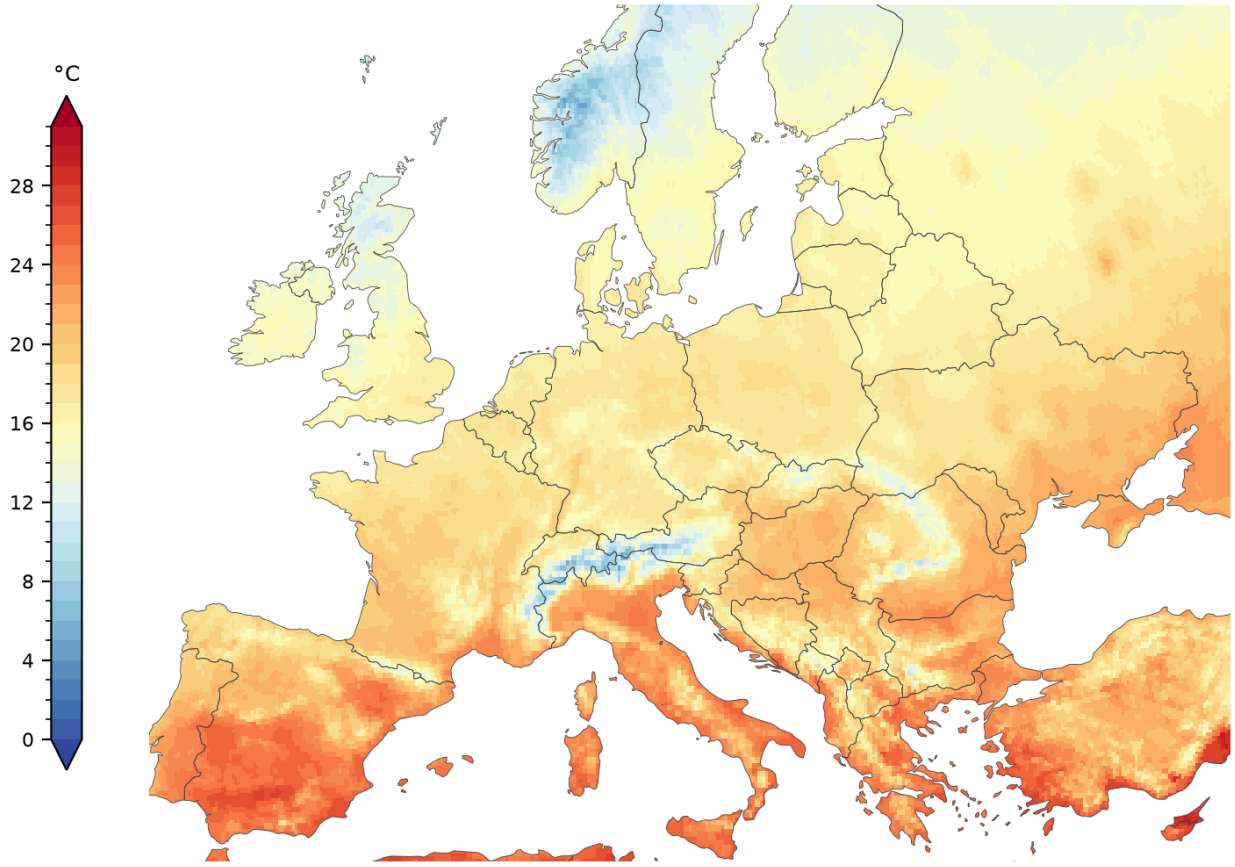
An aerial photo (orthophoto) is a typical multi-band raster: three bands with the red, green and blue values of each cell together form the colour image. Orthophoto: Digitaal Vlaanderen, orthophoto mosaic 2025 (Ghent, Gravensteen).

Continuous and discrete data

Continuous data

The raster model is very suitable for representing **continuous data**. Continuous data are data that have no clear geographic boundaries, or only vague ones, such as air quality, temperature or elevation. If you look at the temperature map of Europe below, you will see that it contains different 'temperature zones'. This is of course a simplification: in reality, the temperature does not jump at a sharp line from exactly 17°C in one zone to 18°C in the next. Phenomena such as temperature change gradually across space and therefore cannot be delimited precisely.

The raster model offers a good way of dealing with this difficulty. Space is systematically divided into equal parts and so simplified into a set of small, indivisible cells. This creates a predefined spatial framework in which information can be recorded. In this simplified representation of space, each cell is given a measured or estimated value that applies to the whole cell.



Mean temperature in August (1970–2000). Each cell (10 × 10 km) has one value. Data: WorldClim 2.1 (Fick & Hijmans 2017), CC BY 4.0; borders: Natural Earth.

Discrete data

Besides continuous data, rasters can also contain **discrete data**. These are data that can be clearly delimited, such as buildings, roads, water bodies… Think, for example, of the buildings and roads you can see on an aerial photo. In raster images, the accuracy of discrete data **depends on the spatial resolution**. The higher the resolution (the smaller the cells), the more detail can be shown and the more accurately boundaries can be drawn.

Simple, but large files

Because the raster model is a very simple data structure, many analyses can be carried out quickly. On the other hand, the files are often large, because information has to be stored for each individual cell.